

Two-Body Quench Dynamics Notes

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1 Introduction

This document is a clean and thorough derivation of the results of the two-body zero-range interaction quench calculations.

The premise here is: we have two atoms in a three-dimensional isotropic harmonic trap interacting with one another via a contact interaction (think zero-range delta function not finite range like billiard balls). We then instantly (much faster than any other timescale) change the value of the interaction strength (i.e. quench in interaction strength) and see what happens. In experiment this is achieved via Feshbach Resonance (a.k.a Fano-Feshbach Resonance). The main reason this is interesting is because zero-range interactions are a commonly used tool and yet things break.

The physical justification of the contact interaction is that the interparticle separation is much greater than the characteristic lengthscale of the interaction (which is typically the van der Waals interaction). This is why this formalism is often used in dilute cold gases, where the particle separation is much greater than the interaction length scale.

You can go into more detail about the exact physics of the interaction. It's called a *s*-wave interaction. The general idea is that when two atoms collide there can be a kind of resonance with

the bounds states of the interatomic potential and this resonance changes the phase of the atoms (phase in the travelling plane wave sense). The greater the resonance the stronger the interaction. The strength of the interaction is parametrised by the s -wave scattering length. The scattering length isn't a real physical length, it's a parameter with units of length that you can link to the phase shift. Also, if collision energy is high other angular momentum channels become important which is why we want the gas to be cold. If you want to get into the real nitty-gritty of the foundations of the contact interaction take a look at my (Alex Kerin) thesis or that of my colleagues Brendan Mulkerin or Chris Bradly. They should be available on the Melbourne University thesis archive.

2 The Two-Body Wavefunction

The most obvious starting place for our investigation into the two-body problem is in the Hamiltonian. The non-interacting problem is well known, so let's start with the interacting problem. The Hamiltonian is given

$$\hat{H}_{2b} = \frac{-\hbar^2}{2m_1} \nabla_1^2 + \frac{-\hbar^2}{2m_2} \nabla_2^2 + \frac{m_1 \omega^2 r_1^2}{2} + \frac{m_2 \omega^2 r_2^2}{2} + \frac{2\pi \hbar^2 a_s}{\mu} \delta^3(|\vec{r}_1 - \vec{r}_2|) \frac{\partial}{\partial(|\vec{r}_1 - \vec{r}_2|)} (|\vec{r}_1 - \vec{r}_2|). \quad (1)$$

where $\mu = \frac{m_1 m_2}{m_1 + m_2}$, $\vec{r}_i$ is the position of the i^{th} particle and the derivative acts on $|\vec{r}_1 - \vec{r}_2|$ times something. People often write the interaction potential in terms of a regularised delta function

$$\hat{V}_{\text{int}} = \frac{4\pi \hbar^2 a_s}{m} \delta^3(|\vec{r}_1 - \vec{r}_2|) \frac{\partial}{\partial(|\vec{r}_1 - \vec{r}_2|)} (|\vec{r}_1 - \vec{r}_2|) = \frac{4\pi \hbar^2 a_s}{m} \delta_{\text{reg}}(\vec{r}_1 - \vec{r}_2).$$

It is confusing that the derivative acts on $|\vec{r}_1 - \vec{r}_2|$ times whatever the interaction terms is acting on but it is the standard way to write it.

This interaction term is called the Fermi-pseudo potential or the Fermi-Huang potential (Huang & Yang, 1957) and was originally used in the context of nuclear physics. It's a regularised delta function rather than an ordinary delta function because using an ordinary delta function leads to non-physical things. Let's briefly look at why we can't just use a normal delta function. Consider a particle in D dimensional free space with a delta function potential at some point.

$$\left(\frac{-\hbar^2}{2m} \nabla^2 - g \delta^D(\vec{r}) \right) \psi = E \psi$$

We're considering a bound state so $g > 0$ and $E < 0$. Fourier transform into momentum space

$$\begin{aligned} E \phi(p) &= \frac{p^2}{2m} \phi(p) - g \psi(0) \\ \phi(p) &= \frac{2mg \psi(0)}{p^2 - 2mE} \end{aligned}$$

the interaction term just filters out the wavefunction at $r = 0$. Doing an inverse fourier transform on $\phi(p)$ and setting $r = 0$ gives us

$$\begin{aligned} \psi(0) &= \int \frac{d^D p}{(2\pi)^D} \phi(p) = \int \frac{d^D p}{(2\pi)^D} \frac{2mg \psi(0)}{p^2 - 2mE} \\ 1 &= \int \frac{d^D p}{(2\pi)^D} \frac{2mg}{p^2 - 2mE} \end{aligned}$$

This integral depends on the dimensionality D $d^D p = \Omega_D p^{D-1} dp$

$$1 = \frac{2mg}{(2\pi)^D} \int \frac{p^{D-1}}{p^2 - 2mE} dp$$

For $D = 1$ this converges, for $D = 2$ this diverges logarithmically, and for $D = 3$ this diverges linearly. This means the delta function potential is only suitable for $D = 1$ and breaks for higher dimension. Below we will use a regularised delta function that does not have this problem.

2.1 Wavefunction Derivation

It's an axiom of quantum mechanics that the wavefunction of system contains all knowable information about the system. If we can get the wavefunction then we have, in principle, gotten everything. In practice getting the information you want out of the wavefunction can be challenging. Still getting the wavefunction is a big deal and we can do it. We'll largely be following Busch et al. (1998) but extended to general mass and with more working shown.

We begin by making a coordinate transform that is going to prove useful

$$\begin{aligned}\vec{R} &= \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2}, \quad r = r_1 - r_2, \\ \vec{r}_1 &= \frac{m_2 \vec{r}}{m_1 + m_2} + \vec{R}, \quad \vec{r}_2 = -\frac{m_1 \vec{r}}{m_1 + m_2} + \vec{R}.\end{aligned}$$

This implies

$$\begin{aligned}\frac{d}{dr_1} &= \frac{\partial R}{\partial r_1} \frac{\partial}{\partial R} + \frac{\partial r}{\partial r_1} \frac{\partial}{\partial r}, \\ \frac{d^2}{dr_1^2} &= \frac{m_1^2}{(m_1 + m_2)^2} \frac{\partial^2}{\partial R^2} + \frac{2m_1}{m_1 + m_2} \frac{\partial}{\partial R} \frac{\partial}{\partial r} + \frac{\partial^2}{\partial r^2}, \\ \frac{d^2}{dr_2^2} &= \frac{m_2^2}{(m_1 + m_2)^2} \frac{\partial^2}{\partial R^2} - \frac{2m_2}{m_1 + m_2} \frac{\partial}{\partial R} \frac{\partial}{\partial r} + \frac{\partial^2}{\partial r^2}.\end{aligned}$$

When you crunch through all the math you get

$$\hat{H}_{2b} = \frac{-\hbar^2}{2(m_1 + m_2)} \nabla_R^2 + \frac{-\hbar^2}{2\mu} \nabla_r^2 + \frac{(m_1 + m_2)\omega^2 R^2}{2} + \frac{\mu\omega^2 r^2}{2} + \frac{2\pi\hbar^2 a_s}{\mu} \delta^3(r) \frac{\partial}{\partial r}(r).$$

It will prove convenient to define

$$\hat{H}_{\text{CoM}} = \frac{-\hbar^2}{2(m_1 + m_2)} \nabla_R^2 + \frac{(m_1 + m_2)\omega^2 R^2}{2} \quad (2)$$

$$\hat{H}_{\text{rel}} = \frac{-\hbar^2}{2\mu} \nabla_r^2 + \frac{\mu\omega^2 r^2}{2} + \frac{2\pi\hbar^2 a_s}{\mu} \delta^3(r) \frac{\partial}{\partial(r)}(r), \quad (3)$$

$$a_{\text{CoM}} = \sqrt{\frac{\hbar}{(m_1 + m_2)\omega}}, \quad a_\mu = \sqrt{\frac{\hbar}{\mu\omega}}.$$

The centre of mass (CoM) part of the wavefunction is a simple harmonic oscillator Hamiltonian but the relative part has the interaction term and is not so simple. Because the Hamiltonian can be separated into CoM and relative parts the wavefunction is can be written

$$\Psi_{2b}(\vec{r}_1, \vec{r}_2) = \psi_{2b}(\vec{r}) \Phi(\vec{R}).$$

The CoM term is just a SHO wavefunction but the relative term is more complicated. We begin by expanding in terms of non-interacting SHO wavefunctions of $l = 0$

$$|\psi_{2b}(\vec{r})\rangle = \sum_{n=0}^{\infty} C_n |\phi_n\rangle.$$

The reason we expand in $l = 0$ states is because states with $l \neq 0$ are zero at $r = 0$, don't experience the contact interaction and therefore don't contribute. The $l \neq 0$ terms go to zero as we'll see later.

It's convenient to define

$$\hat{H}_{\text{osc}} = \frac{-\hbar^2}{2\mu} \nabla_r^2 + \frac{\mu\omega^2 r^2}{2}, \quad \hat{H}_{\text{osc}}\phi_n = E_n\phi_n, \quad \hat{H}_{\text{rel}}\psi_{2b} = E_{\text{rel}}\psi_{2b}, \quad \hat{H}_{\text{rel}} = \hat{H}_{\text{osc}} + \hat{V}_{\text{int}}.$$

So then

$$\begin{aligned} \hat{H}_{\text{rel}}|\psi_{2b}\rangle &= E_{\text{rel}}|\psi_{2b}\rangle = \sum_{k=0}^{\infty} C_k (\hat{H}_{\text{osc}} + \hat{V}_{\text{int}}) |\phi_k\rangle = E_{\text{rel}} \sum_{k=0}^{\infty} C_k |\phi_k\rangle, \\ 0 &= \sum_{k=0}^{\infty} C_k (E_k - E_{\text{rel}}) |\phi_k\rangle + \frac{2\pi\hbar^2 a_s}{\mu} \delta_{\text{reg}}^3(r) |\phi_k\rangle. \end{aligned}$$

Apply some particular state $\langle\phi_n|$

$$\begin{aligned} 0 &= C_n (E_n - E_{\text{rel}}) + \frac{2\pi\hbar^2 a_s}{\mu} \phi_n(0) \left[\frac{\partial}{\partial r} \left(r \sum_{k=0}^{\infty} C_k \phi_k \right) \right]_{r \rightarrow 0}, \\ C_n &= \frac{-\frac{2\pi\hbar^2 a_s}{\mu} \phi_n(0) \left[\frac{\partial}{\partial r} \left(r \sum_{k=0}^{\infty} C_k \phi_k \right) \right]_{r \rightarrow 0}}{E_n - E_{\text{rel}}}. \end{aligned}$$

The subscript $r \rightarrow 0$ means the limit as r goes to 0. It's often written with the subscript. The $l \neq 0$ states don't survive this limit because they are zero at zero. For convenience define A

$$\begin{aligned} A &= -\frac{2\pi\hbar^2 a_s}{\mu} \left[\frac{\partial}{\partial r} \left(r \sum_{k=0}^{\infty} C_k \phi_k \right) \right]_{r \rightarrow 0}, \\ C_n &= \frac{A \phi_n(0)}{E_n - E_{\text{rel}}}, \\ \psi_{2b} &= \sum_{n=0}^{\infty} \frac{A \phi_n(0) \phi_n(r)}{E_n - E_{\text{rel}}}. \end{aligned}$$

The s -wave ($l = 0$) non-interacting wavefunctions are well known (e.g. wikipedia)

$$\phi_n(\vec{r}) = N_n e^{\frac{-r^2}{2a_\mu^2}} L_n^{1/2} \left(\frac{r^2}{a_\mu^2} \right) Y_{00}(\hat{r}), \quad (4)$$

$$N_n = \sqrt{\sqrt{\frac{2(\frac{1}{2a_{\text{rel}}^2})^3}{\pi}} \frac{2^{n+3}n!}{(2n+1)!!}} = \left(\frac{2}{\pi}\right)^{\frac{1}{4}} \left(\frac{1}{2a_{\text{rel}}^2}\right)^{\frac{3}{4}} \sqrt{\frac{2^{n+3}(n!)}{(2n+1)!!}}.$$

where $L_n^{\frac{1}{2}}$ are associated laguerre polynomials and $Y_{00} = 1/2\sqrt{\pi}$ is the zeroth order spherical harmonic. There are a bunch of useful identities to simplify the normalisation

$$\begin{aligned} L_n^\alpha(0) &= \binom{n+\alpha}{n} = \frac{\Gamma(n+\alpha+1)}{\Gamma(n+1)\Gamma(\alpha+1)}, \quad \Gamma(n+\frac{1}{2}) = \frac{(2n-1)!!}{2^n} \sqrt{\pi}, \\ L_n^{\frac{1}{2}}(0) &= \frac{\Gamma(n+\frac{3}{2})}{\Gamma(n+1)\Gamma(\frac{3}{2})} = \frac{(2n+1)!!}{2^n n!}, \quad \phi_n(\vec{r}) = \left(\frac{1}{\pi a_\mu^2}\right)^{\frac{3}{4}} (L_n^{\frac{1}{2}}(0))^{\frac{-1}{2}} e^{\frac{-r^2}{2a_\mu^2}} L_n^{\frac{1}{2}}\left(\frac{r^2}{a_\mu^2}\right). \end{aligned}$$

In analogy with the non-interacting energy $E_n = (2n + 3/2)\hbar\omega$ we define the interacting pseudo quantum number ν such that $E_{\text{rel}} = (2\nu + 3/2)\hbar\omega$. This gives

$$\psi_{2b} = \sum_{n=0}^{\infty} \frac{A\phi_n(0)\phi_n(r)}{E_n - E_{\text{rel}}} = \frac{A}{\hbar\omega} \left(\frac{1}{\pi a_\mu^2}\right)^{\frac{3}{2}} \sum_{n=0}^{\infty} \frac{e^{\frac{-r^2}{2a_\mu^2}} L_n^{\frac{1}{2}}\left(\frac{r^2}{a_\mu^2}\right)}{n - \nu}.$$

From Busch et al. (1998) or e.g. Abramowitz & Stegun (1964) we have the identity

$$\sum_{n=0}^{\infty} \frac{L_n^{\frac{1}{2}}(x^2)}{n - \nu} = \Gamma(-\nu) U\left(-\nu, \frac{3}{2}, x^2\right). \quad (5)$$

Where $U(a, b, z)$ is Kummer's confluent hypergeometric function of the second kind. We then have

$$\begin{aligned} \psi_{2b}(\nu, r) &= \frac{A}{\hbar\omega} \left(\frac{1}{\pi a_\mu^2}\right)^{\frac{3}{2}} e^{\frac{-r^2}{2a_\mu^2}} \Gamma(-\nu) U\left(-\nu, \frac{3}{2}, \frac{r^2}{a_\mu^2}\right), \\ \psi_{2b}(\nu, r) &= N_\nu \Gamma(-\nu) e^{\frac{-r^2}{2a_\mu^2}} U\left(-\nu, \frac{3}{2}, \frac{r^2}{a_\mu^2}\right). \end{aligned} \quad (6)$$

We could roll $\Gamma(-\nu)$ into the normalisation but Busch et al. (1998) didn't and also it ends up being useful later when we want to apply the Laguerre identity Eq. (5).

2.2 Normalisation Derivation

Now that we have the form of the wavefunction the next step is to get the normalization

$$1 = \int_{\text{all space}} |\Psi_{2b}(r_1, r_2)|^2 dr_1 dr_2.$$

The Jacobian between $\vec{r}_1, \vec{r}_2$ and $\vec{r}, \vec{R}$ is one.

$$d\vec{r}_1 d\vec{r}_2 = \det \left(\begin{bmatrix} \frac{d\vec{r}_1}{d\vec{r}} & \frac{d\vec{r}_1}{d\vec{R}} \\ \frac{d\vec{r}_2}{d\vec{r}} & \frac{d\vec{r}_2}{d\vec{R}} \end{bmatrix} \right) d\vec{r} d\vec{R} = \det \left(\begin{bmatrix} \frac{m_2}{m_1 + m_2} & 1 \\ -\frac{m_1}{m_1 + m_2} & 1 \end{bmatrix} \right) d\vec{r} d\vec{R} = d\vec{r} d\vec{R}.$$

The centre of mass wavefunction integrates out to one leaving

$$\begin{aligned} 1 &= \int_{\text{all space}} |\psi_{2b}(r)|^2 dr, \\ &= 4\pi N_\nu^2 \Gamma(-\nu)^2 \int_{r=0}^{r=\infty} r^2 e^{\frac{-r^2}{a_\mu^2}} U\left(-\nu, \frac{3}{2}, \frac{r^2}{a_\mu^2}\right)^2 dr. \end{aligned}$$

Recalling Eq. (5)

$$\begin{aligned} 1 &= 4\pi N_\nu^2 \int_{r=0}^{r=\infty} r^2 e^{\frac{-r^2}{a_\mu^2}} \left(\sum_{n=0}^{\infty} \frac{L_n^{\frac{1}{2}}\left(\frac{r^2}{a_\mu^2}\right)}{n - \nu} \right) \left(\sum_{m=0}^{\infty} \frac{L_m^{\frac{1}{2}}\left(\frac{r^2}{a_\mu^2}\right)}{m - \nu} \right) dr, \\ &= 4\pi N_\nu^2 \sum_{n,m=0}^{\infty} \frac{1}{(n - \nu)(m - \nu)} \int_{r=0}^{r=\infty} r^2 e^{\frac{-r^2}{a_\mu^2}} L_n^{\frac{1}{2}}\left(\frac{r^2}{a_\mu^2}\right) L_m^{\frac{1}{2}}\left(\frac{r^2}{a_\mu^2}\right) dr. \end{aligned}$$

The trick is now to evaluate that integral

$$\begin{aligned}
I &= \int_{r=0}^{r=\infty} r^2 e^{\frac{-r^2}{a_\mu^2}} L_n^{\frac{1}{2}}\left(\frac{r^2}{a_\mu^2}\right) L_m^{\frac{1}{2}}\left(\frac{r^2}{a_\mu^2}\right) dr \\
x &= \frac{r^2}{a_\mu^2}, \quad dr = \frac{dr}{dx} dx = \frac{a_\mu^2}{2r} dx \\
I &= \frac{a_\mu^3}{2} \int_{x=0}^{x=\infty} \sqrt{x} e^{-x} L_n^{\frac{1}{2}}(x) L_m^{\frac{1}{2}}(x) dx.
\end{aligned}$$

From Srivastava et al. (2003) or e.g. Gradshteyn & Ryzhik (2014).

$$\int_{x=0}^{x=\infty} x^\alpha e^{-x} L_n^\alpha(x) L_m^\alpha(x) dx = \binom{n+\alpha}{n} \Gamma(\alpha+1) \delta_{m,n}, = \frac{\Gamma(n+\alpha+1)}{\Gamma(n+1)} \delta_{m,n}$$

$\mathbb{R}(\alpha) > -1 \quad m, n \in \mathbb{Z}_{\geq 0}.$

Circling back around we have

$$\begin{aligned}
I &= \frac{a_\mu^3}{2} \frac{\Gamma(n+3/2)}{\Gamma(n+1)} \delta_{m,n}, \\
1 &= 2\pi N_\nu^2 a_\mu^3 \sum_{n=0}^{\infty} \frac{\Gamma(n+\frac{3}{2})}{\Gamma(n+1)(n-\nu)^2}.
\end{aligned}$$

We define the sum as $Z(\nu)$ for convenience and according to wolfram alpha/mathematica that sum is

$$Z(\nu) = \sum_{n=0}^{\infty} \frac{\Gamma(n+\frac{3}{2})}{\Gamma(n+1)(n-\nu)^2}, \tag{7}$$

$$Z(\nu) = \frac{\pi \Gamma(1-\nu) (\psi^{(0)}(-\nu - \frac{1}{2}) - \psi^{(0)}(-\nu))}{\nu \Gamma(-\nu - \frac{1}{2})}, \tag{8}$$

where $\psi^{(n)}(x)$ is the polygamma function, the n^{th} of the digamma function which is the logarithmic derivative of the Gamma function. Putting it together

$$\begin{aligned}
\psi_{2b}(\nu, r) &= N_\nu \Gamma(-\nu) e^{\frac{r^2}{2a_\mu^2}} U\left(-\nu, \frac{3}{2}, \frac{r^2}{a_\mu^2}\right), \\
N_\nu &= (2\pi a_\mu^3 Z(\nu))^{-\frac{1}{2}}.
\end{aligned} \tag{9}$$

And so we have the interacting two-body wavefunction.

3 The Energy Spectrum and Unitary/Non-Interacting Limits

This is excellent, we now have the interacting two-body wavefunction. But what are the values of ν and how do they relate to a_s ? First, recall that

$$\begin{aligned}
0 &= C_n(E_n - E_{rel}) + \frac{2\pi\hbar^2 a_s}{\mu} \phi_n(0) \left[\frac{\partial}{\partial r} \left(r \sum_{m=0}^{\infty} C_m \phi_m \right) \right]_{r \rightarrow 0}, \\
C_n &= \frac{A \phi_n(0)}{E_n - E_{rel}}.
\end{aligned}$$

So then

$$\begin{aligned}
0 &= A\phi_n(0) + \frac{2\pi\hbar^2 a_s}{\mu} \phi_n(0) \left[\frac{\partial}{\partial r} \left(r \sum_{m=0}^{\infty} \frac{A\phi_m(0)\phi_m}{E_n - E_{rel}} \right) \right]_{r \rightarrow 0}, \\
\frac{-\mu}{2\pi\hbar^2 a_s} &= \left[\frac{\partial}{\partial r} \left(r \sum_{n=0}^{\infty} \frac{\phi_m(0)\phi_m}{E_n - E_{rel}} \right) \right]_{r \rightarrow 0}, \\
&= \left[\frac{\partial}{\partial r} \left(r \sum_{n=0}^{\infty} \frac{(\pi a_\mu^2)^{\frac{-3}{2}} e^{-\frac{r^2}{2a_\mu^2}} L_n^{\frac{1}{2}} \left(\frac{r^2}{a_\mu^2} \right)}{2\hbar\omega(n - \nu)} \right) \right]_{r \rightarrow 0}, \\
&= \frac{1}{(\pi a_\mu^2)^{\frac{3}{2}} (2\hbar\omega)} \left[\frac{\partial}{\partial r} \left(r \sum_{n=0}^{\infty} \frac{e^{-\frac{r^2}{2a_\mu^2}} L_n^{\frac{1}{2}} \left(\frac{r^2}{a_\mu^2} \right)}{(n - \nu)} \right) \right]_{r \rightarrow 0}, \\
\frac{-\sqrt{\pi} a_\mu}{a_s} &= \left[\frac{\partial}{\partial r} \left(r \sum_{n=0}^{\infty} \frac{e^{-\frac{r^2}{2a_\mu^2}} L_n^{\frac{1}{2}} \left(\frac{r^2}{a_\mu^2} \right)}{(n - \nu)} \right) \right]_{r \rightarrow 0}.
\end{aligned}$$

Recalling the connection between Laguerre polynomials and Tricomi's function Eq. (5)

$$\frac{-\sqrt{\pi} a_{rel}}{a_s} = \left[\frac{\partial}{\partial r} \left(r e^{-\frac{r^2}{2a_\mu^2}} \Gamma(-\nu) U(-\nu, \frac{3}{2}, \frac{r^2}{a_\mu^2}) \right) \right]_{r \rightarrow 0}.$$

For small z $U(a, b, z)$ behaves as (Abramowitz & Stegun, 1964)

$$U(a, b, z) \approx \frac{\Gamma(b-1)}{\Gamma(a)} z^{1-b} + \begin{cases} \mathcal{O}(|z|^{\text{Re}(b-2)}), & \text{if } \text{Re}(b) \geq 2, b \neq 2 \\ \mathcal{O}(\ln(z)), & \text{if } b = 2 \\ \mathcal{O}(1), & \text{if } 1 < \text{Re}(b) < 2 \end{cases} \quad \text{for small } z.$$

Also note that according to Busch et al. (1998) we we have

$$U(-\nu, \frac{3}{2}, \frac{r^2}{a_\mu^2}) = \frac{\sqrt{\pi}}{\Gamma(-\nu)} \frac{a_\mu}{r} - \frac{2\sqrt{\pi}}{\Gamma(-\nu - \frac{1}{2})} + \mathcal{O}\left(\frac{r}{a_\mu}\right).$$

I'm not sure how they get the precise value of the $\mathcal{O}(1)$ term. In any case, only one term survives the limit as $r \rightarrow 0$ so

$$\frac{a_\mu}{a_s} = \frac{2\Gamma(-\nu)}{\Gamma(-\nu - \frac{1}{2})}. \quad (10)$$

Now we have a transcendental equation that relates ν and a_s and can this build the energy spectrum. As a side note, I've seen people define lengthscale $a_m = \sqrt{\hbar/m\omega}$ not $a_\mu = \sqrt{\hbar/\mu\omega}$ which introduces a factor of $\sqrt{2}$ which causes Eq. (10) to have a factor of $\sqrt{2}$ rather than 2 in the numerator.

Solving Eq. (10) numerically gives a spectrum like Fig. 1. Note that as the energy increases the gradient near a_s becomes steeper and steeper. Now while the bound state only exists for positive a_s the literature often refers to $a_s < 0$ as the attractive regime because the rest of the energies decrease compared to the non-interacting energies.

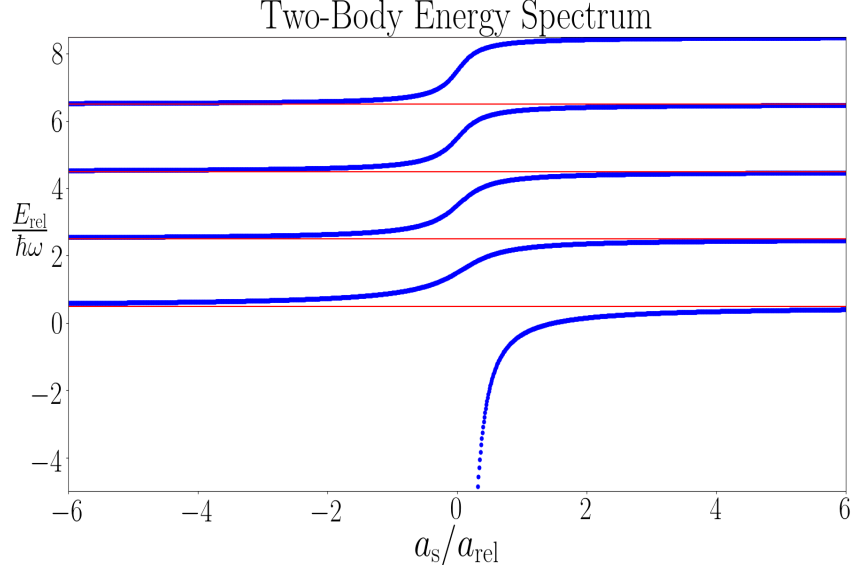


Figure 1: Eigenenergies of $\hat{H}_{\text{rel}}$ as a function of the s-wave scattering length (blue dots). The red lines correspond to $E_{\text{rel}} = (2n + 1/2)\hbar\omega$ with $n \in \mathbb{Z}_{\geq 0}$, the eigenenergy of the relative Hamiltonian in the unitary limit ($a_s/a_\mu \rightarrow \pm\infty$). At $a_s/a_\mu \rightarrow 0$ we have $E_{\text{rel}} = (2n + 3/2)\hbar\omega$ with $n \in \mathbb{Z}_{\geq 0}$, which is the standard non-interacting energy spectrum.

Because it is a transcendental we can't get an exact spectrum. However, there are two cases we can get exact results, the non-interacting and strongly interacting (unitary) limits. The strongly interacting, commonly called the unitary limit, is when $a_s \rightarrow \infty$, i.e. When the s -wave scattering length becomes much larger than any other lengthscale of the system. Naturally, in the weakly interacting limit $a_s = 0$. In these limits we have

$$\begin{aligned} \text{non-interacting} : \nu &= n, \quad n \in \mathbb{Z}_{\geq 0}, \\ \text{strongly interacting} : \nu &= j - \frac{1}{2}, \quad j \in \mathbb{Z}_{\geq 0}. \end{aligned}$$

Worth noting that since in the non-interacting case the ν s are integers you can't use Eq. (6) since a Γ function of a negative integer is undefined. But you can just use the normal SHO wavefunctions

In the unitary case all of the ν s are nice half integers. This lets us make some simplifications. The most notably the normalisation. As per Eq. (9)

$$\begin{aligned} N_\nu &= (2\pi a_\mu^3 Z(\nu))^{-1/2}, \\ Z(\nu) &= \frac{\pi\Gamma(1-\nu)(\psi^{(0)}(-\nu - \frac{1}{2}) - \psi^{(0)}(-\nu))}{\nu\Gamma(-\nu - \frac{1}{2})}. \end{aligned}$$

I'm going to use the notation ν_j to denote the j^{th} excited state and in the unitary limit $\nu_j = j - 1/2$.

$$\begin{aligned}
Z(\nu_j) &= \frac{\pi\Gamma(1-\nu_j)(\psi^{(0)}(-\nu_j-\frac{1}{2})-\psi^{(0)}(-\nu_j))}{\nu_j\Gamma(-\nu_j-\frac{1}{2})}, \\
Z(j-\frac{1}{2}) &= \frac{\pi\Gamma(\frac{3}{2}-j)(\psi^{(0)}(-j)-\psi^{(0)}(-j+1/2))}{(j-\frac{1}{2})\Gamma(-j)}, \\
&= \frac{\pi\Gamma(\frac{3}{2}-j)}{j-\frac{1}{2}} \frac{\psi^{(0)}(-j)-\psi^{(0)}(-j+1/2)}{\Gamma(-j)}.
\end{aligned}$$

Strictly it's a singularity here for $j \in \mathbb{Z}_{\geq 0}$ but a limit does exist,

$$\lim_{x \rightarrow -N} \frac{\Gamma(x)}{\psi^{(0)}(x)} = \frac{(-1)^{1-N}}{N!}, \quad \lim_{x \rightarrow -N} \frac{\Gamma(x)}{\psi^{(0)}(x+\frac{1}{2})} = \pm\infty, \quad N \in \mathbb{N}.$$

$+\infty$ for even N and $-\infty$ for odd N . So we have

$$\begin{aligned}
Z(j-\frac{1}{2}) &= \frac{\pi\Gamma(\frac{3}{2}-j)}{j-\frac{1}{2}} \frac{j!}{(-1)^{1-j}}, \\
&= \pi\Gamma(\frac{1}{2}-j)(-1)^j j!, \\
\text{well known identity : } \pi(-1)^j &= \Gamma(\frac{1}{2}-j)\Gamma(\frac{1}{2}+j), \\
Z(j-\frac{1}{2}) &= \pi^2 \frac{\Gamma(1+j)}{\Gamma(\frac{1}{2}+j)}.
\end{aligned}$$

So in the unitary case the normalisation is

$$N_{\nu_j} = \left(2\pi^3 \frac{\Gamma(1+j)}{\Gamma(\frac{1}{2}+j)}\right)^{-\frac{1}{2}}.$$

There are also some simplifications to be made in the non-interacting limit also. From e.g. wikipedia we have the identity

$$L_n^\alpha(x) = \frac{(-1)^n}{n!} U(-n, \alpha+1, x).$$

4 Ramsey Signal

Above in Sec. 2 we get the interacting wavefunction and the energy spectra. Now we want to talk about quench dynamics. Again, we're quenching in a_s . The way you actually do that in experiment is by exploiting a phenomenon known as Feshbach resonance. Broadly, you apply an external magnetic field that affects the atom's internal states which affects how strongly they interact with one another.

The Ramsey signal is the overlap of the pre- and post-quench wavefunction states

$$\begin{aligned}
S(t) &= \langle \Psi_{\text{pre}}(t) | \Psi_{\text{post}}(t) \rangle, \\
&= \langle \Psi_{\text{pre}}(0) | e^{i\hat{H}_{\text{pre}}t\hbar} e^{-i\hat{H}_{\text{post}}t\hbar} | \Psi_{\text{post}}(0) \rangle.
\end{aligned} \tag{11}$$

Where Ψ_{pre} is the pre-quench wavefunction and Ψ_{post} is the post-quench wavefunction. Similarly $\hat{H}_{\text{pre}}$ is the pre-quench Hamiltonian and $\hat{H}_{\text{post}}$ is the post-quench Hamiltonian. The magnitude of

the Ramsey signal is sometimes called the contrast and it's related to the fidelity. The Ramsey signal can be measured in experiment Cetina et al. (2016).

We proceed by inserting a complete set of post-quench eigenstates ϕ_n

$$S(t) = \sum_{n=0}^{\infty} |\langle \psi_0(0) | \phi_n \rangle|^2 e^{-i(E_n - E_{\text{pre}})t/\hbar}.$$

Where E_{pre} is the energy of the pre-quench state and E_n are eigenenergies of the post-quench Hamiltonian. Now it's a matter of calculating the overlap coefficients. It's worth mentioning that while we can choose the initial state we don't have control over what the final state is. We can quench from e.g. the 3rd excited state specifically but not to any particular state.

There are three possible systems we can quench between, non-interacting, general a_s , or unitary. Let's start with the interacting to interacting case. Here ν_i corresponds to the initial state and ν_j to the post-quench eigenstates.

$$\begin{aligned} S(t) &= \sum_{j=0}^{\infty} |\langle \psi_i | \phi_{\nu_j} \rangle|^2 e^{-2i(\nu_j - \nu_i)\omega t}, \\ \psi_i(\nu_i, r) &= N_{\nu_i} e^{\frac{r^2}{2a_\mu^2}} \Gamma(-\nu_i) U\left(-\nu_i, \frac{3}{2}, \frac{r^2}{a_\mu^2}\right), \\ \phi_{\nu_j}(\nu_j, r) &= N_{\nu_j} e^{\frac{r^2}{2a_\mu^2}} \Gamma(-\nu_j) U\left(-\nu_j, \frac{3}{2}, \frac{r^2}{a_\mu^2}\right), \\ \langle \psi_i | \phi_{\nu_j} \rangle &= 4\pi N_{\nu_i} N_{\nu_j} \Gamma(-\nu_i) \Gamma(-\nu_j) \int_{r=0}^{r=\infty} r^2 e^{\frac{r^2}{a_\mu^2}} U\left(-\nu_i, \frac{3}{2}, \frac{r^2}{a_\mu^2}\right) U\left(-\nu_j, \frac{3}{2}, \frac{r^2}{a_\mu^2}\right) dr. \end{aligned}$$

Recall Eq. (5), the expansion of U into Laguerre polynomials

$$\begin{aligned} \langle \psi_i | \phi_j \rangle &= 4\pi N_{\nu_i} N_{\nu_j} \int_{r=0}^{r=\infty} r^2 e^{-r^2/a_\mu^2} \left(\sum_{n=0}^{\infty} \frac{L_n^{\frac{1}{2}}(\frac{r^2}{a_\mu^2})}{n - \nu_i} \right) \left(\sum_{m=0}^{\infty} \frac{L_m^{\frac{1}{2}}(\frac{r^2}{a_\mu^2})}{m - \nu_j} \right) dr, \\ \text{defining :} \quad x &= \frac{r^2}{a_\mu^2}, \quad dr = \frac{dr}{dx} dx = \frac{a_\mu^2}{2r} dx, \\ \langle \psi_i | \phi_j \rangle &= \sum_{n,m=0}^{\infty} \frac{2\pi N_{\nu_i} N_{\nu_j} a_\mu^3}{(n - \nu_i)(m - \nu_j)} \int_{x=0}^{x=\infty} \sqrt{x} e^{-x} L_n^{\frac{1}{2}}(x) L_m^{\frac{1}{2}}(x) dx. \end{aligned}$$

The standard Laguerre polynomial orthogonality condition is

$$\int_0^\infty \sqrt{x} e^{-x} L_n^{\frac{1}{2}}(x) L_m^{\frac{1}{2}}(x) dx = \binom{n+1/2}{n} \Gamma\left(\frac{3}{2}\right) \delta_{n,m} = \frac{\Gamma(n + \frac{3}{2})}{\Gamma(n+1)} \delta_{n,m}. \quad (12)$$

So we have

$$\begin{aligned}
\langle \psi_{\nu_i} | \phi_j \rangle &= 2\pi N_{\nu_i} N_{\nu_j} a_\mu^3 \sum_{n=0}^{\infty} \frac{\Gamma(n + \frac{3}{2})}{(n - \nu_i)(n - \nu_j)\Gamma(n + 1)}, \\
\sum_{n=0}^{\infty} \frac{\Gamma(n + \frac{3}{2})}{(n - \nu_i)(n - \nu_j)\Gamma(n + 1)} &= \frac{\sqrt{\pi}}{2\nu_i\nu_j} {}_3F_2\left(\frac{3}{2}, -\nu_j, -\nu_i; 1 - \nu_j, 1 - \nu_i; 1\right), \text{ (from Wolfram Alpha)} \\
\langle \psi_{\nu_i} | \phi_j \rangle &= 2\pi N_{\nu_i} N_{\nu_j} a_\mu^3 \frac{\sqrt{\pi}}{2\nu_i\nu_j} {}_3F_2\left(\frac{3}{2}, -\nu_j, -\nu_i; 1 - \nu_j, 1 - \nu_i; 1\right), \\
\langle \psi_{\nu_i} | \phi_j \rangle &= 2\pi \frac{\sqrt{\pi} a_\mu^3}{2\nu_i\nu_j} (2\pi a_\mu^3 Z(\nu_i))^{-\frac{1}{2}} (2\pi a_\mu^3 Z(\nu_j))^{-\frac{1}{2}} \\
&\quad \times {}_3F_2\left(\frac{3}{2}, -\nu_j, -\nu_i; 1 - \nu_j, 1 - \nu_i; 1\right), \\
\langle \psi_{\nu_i} | \phi_j \rangle &= \frac{\sqrt{\pi}}{2\nu_i\nu_j} \frac{1}{\sqrt{Z(\nu_i)Z(\nu_j)}} {}_3F_2\left(\frac{3}{2}, -\nu_j, -\nu_i; 1 - \nu_j, 1 - \nu_i; 1\right). \tag{13}
\end{aligned}$$

Now that we have the overlaps between wavefunctions we can calculate the Ramsey signal for quenches between non-zero finite a_s .

Now let's move on to the non-interacting to interacting quench. This gives us the interacting to non-interacting quench also. Recall the interacting and non-interacting wavefunctions are respectively

$$\begin{aligned}
\phi_{\nu_j}(r) &= N_{\nu_j} \Gamma(-\nu_j) e^{\frac{r^2}{2a_\mu^2}} U\left(-\nu_j, \frac{3}{2}, \frac{r^2}{a_\mu^2}\right), \\
\psi_n(\vec{r}) &= N_n e^{\frac{-r^2}{2a_\mu^2}} L_n^{1/2}\left(\frac{r^2}{a_\mu^2}\right) Y_{00}(\hat{r}).
\end{aligned}$$

The key thing is the overlap, which is

$$\begin{aligned}
\langle \phi_{\nu_j} | \psi_n \rangle &= 2\sqrt{\pi} N_{\nu_j} N_n \Gamma(-\nu_j) \int_{r=0}^{r=\infty} r^2 e^{\frac{r^2}{a_\mu^2}} U\left(-\nu_j, \frac{3}{2}, \frac{r^2}{a_\mu^2}\right) L_n^{1/2}\left(\frac{r^2}{a_\mu^2}\right) dr, \\
&= 2\sqrt{\pi} N_{\nu_j} N_n \int_{k=0}^{\infty} \frac{L_k^{\frac{1}{2}}\left(\frac{r^2}{a_\mu^2}\right)}{k - \nu_j} L_n^{\frac{1}{2}}\left(\frac{r^2}{a_\mu^2}\right) dr, \\
&= \frac{2\sqrt{\pi} N_{\nu_j} N_n}{n - \nu_j} \binom{n + 1/2}{n} \Gamma\left(\frac{3}{2}\right), \\
&= \frac{2\sqrt{\pi} N_{\nu_j} N_n}{n - \nu_j} \frac{\Gamma(n + \frac{3}{2})}{\Gamma(n + 1)} \Gamma\left(\frac{3}{2}\right) a_\mu^3, \\
&= \frac{2\sqrt{\pi}}{n - \nu_j} \frac{\Gamma(n + \frac{3}{2})}{\Gamma(n + 1)} \Gamma\left(\frac{3}{2}\right) \left(\frac{1}{\pi a_\mu^2}\right)^{3/4} \\
&\quad \times \left(\frac{\Gamma(n + \frac{3}{2})}{\Gamma(n + 1)\Gamma(\frac{3}{2})}\right)^{-1/2} (2\pi a_\mu^3 Z(\nu_j))^{-1/2} a_\mu^3, \\
&= \frac{2\sqrt{\pi}}{n - \nu_j} \frac{\Gamma(n + \frac{3}{2})}{\Gamma(n + 1)} \sqrt{\Gamma\left(\frac{3}{2}\right) \left(\frac{1}{\pi}\right)^{3/4} \left(\frac{\Gamma(n + \frac{3}{2})}{\Gamma(n + 1)}\right)^{-1/2}} (2\pi Z(\nu_j))^{-1/2}, \\
&= \frac{1}{(n - \nu_j)} \sqrt{\frac{\Gamma(n + \frac{3}{2})}{\Gamma(n + 1)}} (Z(\nu_j))^{-1/2}. \tag{14}
\end{aligned}$$

In the unitary case we have

$$\begin{aligned} Z(\nu_j) &= \pi^2 \frac{\Gamma(1+j)}{\Gamma(\frac{1}{2}+j)}, \\ \langle \phi_{\nu_j} | \psi_n \rangle &= \frac{\sqrt{n+\frac{1}{2}}}{\pi(n-j+\frac{1}{2})} \sqrt{\frac{\Gamma(\frac{1}{2}+n)\Gamma(\frac{1}{2}+j)}{\Gamma(1+n)\Gamma(1+j)}}. \end{aligned} \quad (15)$$

The quenches between the limits are the most interesting ones. Let's look at the non-interacting to unitary (forwards) quench first.

$$\begin{aligned} S(t) &= \sum_{j=0}^{\infty} |\langle \psi_n | \phi_{\nu_j} \rangle|^2 e^{-i(E_j - E_n)t\omega}, \\ &= \frac{\Gamma(n+\frac{3}{2})}{\pi^2 \Gamma(n+1)} \sum_{j=0}^{\infty} \frac{\Gamma(\frac{1}{2}+j)}{(n-j+\frac{1}{2})^2 \Gamma(j+1)} e^{-2i(j-\frac{1}{2}-n)t\omega}, \\ &= \left[\frac{\Gamma(n+\frac{3}{2})}{\pi^2 \Gamma(n+1)} \sum_{j=0}^{\infty} \frac{\Gamma(\frac{1}{2}+j)}{(n-j+\frac{1}{2})^2 \Gamma(j+1)} e^{-2i(j-\frac{1}{2})t\omega} \right] e^{2in\omega t}, \\ &= \left[\frac{2}{\pi} e^{i\omega t} + \frac{1}{\pi} e^{-i\omega t} + \frac{1}{12\pi} e^{-3i\omega t} + \frac{1}{40\pi} e^{-5i\omega t} + \dots \right] e^{2in\omega t} \end{aligned}$$

According to Mathematica

$$S(t) = \frac{4e^{i(2n+1)\omega t} \Gamma(n+\frac{3}{2}) {}_3F_2(\frac{1}{2}, \frac{-1}{2} - n, \frac{-1}{2} - n; \frac{1}{2} - n, \frac{1}{2} - n; e^{-2i\omega t})}{(1+2n)^2 \pi^{\frac{3}{2}} \Gamma(1+n)}. \quad (16)$$

If the initial non-interacting state is the ground state ($n=0$) then this simplifies to

$$S(t) = \frac{2}{\pi} \left(e^{i\omega t} \sqrt{1 - e^{-2i\omega t}} + \arcsin(e^{-i\omega t}) \right),$$

and for short times

$$S(t) \approx 1 + \frac{11}{3\sqrt{2}\pi} e^{-i\pi/4} (\omega t)^{3/2} \approx 1 + 0.825 \dots (\omega t)^{3/2}.$$

This short time behaviour is really similar to Parish & Levinsen (2016) but they have 0.829... not 0.825... and it depends on mass. But their problem is very different, they have an untrapped system with an impurity in a sea. Nonetheless they get very similar results.

Now let's look at the unitary to non-interacting (backwards or reverse) quench

$$\begin{aligned} S(t) &= \sum_{n=0}^{\infty} |\langle \psi_n | \phi_{\nu_j} \rangle|^2 e^{-i(E_j - E_n)t\omega}, \\ S(t) &= \frac{\Gamma(j+\frac{1}{2})}{\pi^2 \Gamma(j+1)} \sum_{n'=0}^{\infty} \frac{1}{n-j+\frac{3}{2}} \frac{\Gamma(n+\frac{1}{2})}{\Gamma(n+1)} e^{-2i(n-j+\frac{1}{2})t\omega}, \\ S(t) &= \frac{2e^{-i(1-2j)t\omega} \Gamma(j+\frac{1}{2})}{(2j-1)^2 \pi^{\frac{3}{2}} \Gamma(j+1)} {}_3F_2\left(\frac{3}{2}, \frac{1}{2} - j, \frac{1}{2} - j; \frac{3}{2} - j, \frac{3}{2} - j; e^{-2it\omega}\right). \end{aligned} \quad (17)$$

If we start in the ground state then

$$S(t) = \frac{2}{\pi} \arcsin(e^{-i\omega t}) \quad (18)$$

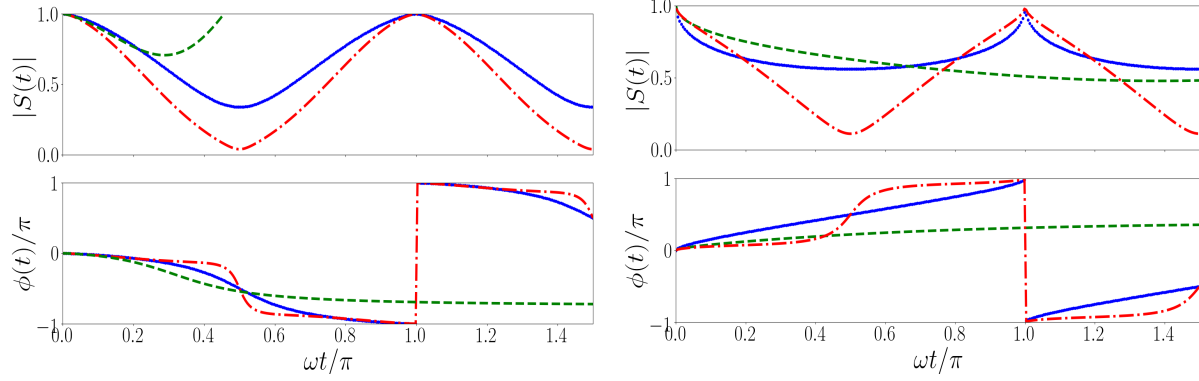


Figure 2: Left panel is non-interacting to unitary quench and right panel is unitary to non-interacting quench. Blue is the exact ground state result. Red is the when the initial state is the first excited state and green is the early time expansion for the ground state. The Ramsey signal is parametrized as $S(t) = |S(t)|e^{-i\phi(t)}$.

4.1 Measuring the Ramsey Signal

But how do we actually measure the Ramsey signal? Using Ramsey interferometry techniques you can relate the Ramsey signal to the probabilities the system is in a particular state.

Consider our two distinguishable particles 1 and 2. Suppose that particle 1 can only be in one state $|1\rangle$ but particle 2 has two states $|2 \uparrow\rangle$ and $|2 \downarrow\rangle$, where $|2 \downarrow\rangle$ (say) doesn't interact with $|1\rangle$ and $|2 \uparrow\rangle$ does interact and can be controlled via Feshbach resonance. In other words, suppose the resonance peaks of $|2 \uparrow\rangle$ and $|2 \downarrow\rangle$ are far away from one another. In experiment this has been achieved by using two different species of atom e.g. Cetina et al. (2016) used Potassium and Lithium. The $|2 \downarrow\rangle$ and $|2 \uparrow\rangle$ would correspond to different excited/spin states of the one of the atomic species. Following Parish & Levinsen (2016) and Cetina et al. (2016) you can obtain the Ramsey signal from the following procedure.

Initialise the system totally in the spin-down state

$$|\psi(0)\rangle = (a|2 \uparrow\rangle + b|2 \downarrow\rangle) \otimes |1\rangle, \quad a^2 + b^2 = 1.$$

Then apply an $\pi/2$ pulse ($U_{\pi/2}^\varphi$), let it evolve for time t ($T(t)$) then hit with another $\pi/2$ pulse. In general there will be a phase difference between the two $\pi/2$ pulses. The operators are

$$U_{\pi/2}^\varphi = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -ie^{i\varphi} \\ -ie^{-i\varphi} & 1 \end{bmatrix}, \quad T(t) = \begin{bmatrix} e^{-i\hat{H}_f t/\hbar} & 0 \\ 0 & e^{-i\hat{H}_0 t/\hbar} \end{bmatrix}.$$

Applying the $\pi/2$ pulses and time evolution operators to the general initial state $|\psi(t)\rangle$ (saying for convenience that the first $\pi/2$ pulse has zero phase and the second has φ) we have

$$|\psi(t)\rangle = U_{\pi/2}^\varphi T(t) U_{\pi/2}^{\varphi=0} |\psi(0)\rangle = U_{\pi/2}^\varphi T(t) U_{\pi/2}^{\varphi=0} \begin{bmatrix} a \\ b \end{bmatrix} |1\rangle.$$

$$|\psi(t)\rangle = \frac{1}{2} \begin{bmatrix} a(e^{-i\hat{H}_f t/\hbar} - e^{i\varphi} e^{-i\hat{H}_0 t/\hbar}) - ib(e^{-i\hat{H}_f t/\hbar} + e^{i\varphi} e^{-i\hat{H}_0 t/\hbar}) \\ -ia(e^{-i\varphi} e^{-i\hat{H}_f t/\hbar} + e^{-i\hat{H}_0 t/\hbar}) + b(-e^{-i\varphi} e^{-i\hat{H}_f t/\hbar} + e^{-i\hat{H}_0 t/\hbar}) \end{bmatrix}.$$

The upper and lower entries in the above matrix are the coefficients of the $|2 \uparrow\rangle \otimes |1\rangle$, and $|2 \downarrow\rangle \otimes |1\rangle$ states after the pulse schema is applies. The probabilities are then.

$$\begin{aligned}
Pr(\uparrow) &= \langle \psi(0) | \frac{1}{4} (a(e^{-i\hat{H}_f t/\hbar} - e^{i\varphi} e^{-i\hat{H}_0 t/\hbar}) - ib(e^{-i\hat{H}_f t/\hbar} + e^{i\varphi} e^{-i\hat{H}_0 t/\hbar})) \\
&\times (a(e^{i\hat{H}_f t/\hbar} - e^{-i\varphi} e^{i\hat{H}_0 t/\hbar}) + ib(e^{i\hat{H}_f t/\hbar} + e^{-i\varphi} e^{i\hat{H}_0 t/\hbar})) | \psi(0) \rangle \\
&= \langle \psi(0) | \frac{1}{4} (a^2(2 - e^{i\varphi} e^{-i(H_0 - H_f)t/\hbar} - e^{-i\varphi} e^{i(H_0 - H_f)t/\hbar}) \\
&\quad + iab(e^{-i\varphi} e^{-i(H_f - H_0)t/\hbar} - e^{i\varphi} e^{i(H_f - H_0)t/\hbar}) \\
&\quad - iab(e^{i\varphi} e^{i(H_f - H_0)t/\hbar} - e^{-i\varphi} e^{-i(H_f - H_0)t/\hbar}) \\
&\quad + b^2(2 + e^{-i\varphi} e^{-i(H_f - H_0)t/\hbar} + e^{i\varphi} e^{i(H_f - H_0)t/\hbar})) | \psi(0) \rangle \\
&= \langle \psi(0) | \frac{1}{2} ((a^2 + b^2) + (b^2 - a^2) \text{Re}[e^{i\varphi} e^{-i(\hat{H}_f - \hat{H}_0)t/\hbar}]) \\
&\quad - ab \text{Im}[e^{-i\varphi} e^{-i(\hat{H}_f - \hat{H}_0)t/\hbar}]) | \psi(0) \rangle \\
Pr(\downarrow) &= \langle \psi(0) | \frac{1}{4} (-ia(e^{-i\varphi} e^{-i\hat{H}_f t/\hbar} + e^{-i\hat{H}_0 t/\hbar}) + b(-e^{-i\varphi} e^{-i\hat{H}_f t/\hbar} + e^{-i\hat{H}_0 t/\hbar})) \\
&\times (ia(e^{i\varphi} e^{i\hat{H}_f t/\hbar} + e^{i\hat{H}_0 t/\hbar}) + b(-e^{i\varphi} e^{i\hat{H}_f t/\hbar} + e^{i\hat{H}_0 t/\hbar})) | \psi(0) \rangle \\
&= \langle \psi(0) | \frac{1}{4} (a^2(2 + e^{-i\varphi} e^{-i(H_f - H_0)t/\hbar} + e^{i\varphi} e^{i(H_f - H_0)t/\hbar}) \\
&\quad - iab(e^{-i\varphi} e^{-i(H_f - H_0)t/\hbar} - e^{i\varphi} e^{i(H_f - H_0)t/\hbar}) \\
&\quad + iab(-e^{-i\varphi} e^{-i(H_f - H_0)t/\hbar} + e^{i\varphi} e^{i(H_f - H_0)t/\hbar}) \\
&\quad + b^2(2 - e^{-i\varphi} e^{-i(H_f - H_0)t/\hbar} - e^{i\varphi} e^{i(H_f - H_0)t/\hbar})) | \psi(0) \rangle \\
&= \langle \psi(0) | \frac{1}{2} ((a^2 + b^2) + (a^2 - b^2) \text{Re}[e^{i\varphi} e^{i(H_f - H_0)t/\hbar}]) \\
&\quad + ab \text{Im}[e^{-i\varphi} e^{-i(H_f - H_0)t/\hbar}]) | \psi(0) \rangle \\
P_{\uparrow} - P_{\downarrow} &= \langle \psi(0) | (b^2 - a^2) \text{Re}[e^{-i\varphi} e^{-i(H_f - H_0)t/\hbar}] - 2ab \text{Im}[e^{-i\varphi} e^{-i(H_f - H_0)t/\hbar}] | \psi(0) \rangle \\
P_{\uparrow} - P_{\downarrow} &= (b^2 - a^2) \text{Re}[e^{-i\varphi} S(t)] - 2ab \text{Im}[e^{-i\varphi} S(t)].
\end{aligned}$$

This then means experimentally we can measure the Ramsey signal from the differency of probabilities of finding the second atom in it's initial or final state, provided that we know the phase difference (φ) of the two $\pi/2$ pulses and the initial a and b values. And we do know what those are because they come from the experimental set up.

5 Particle Separation

The Ramsey signal is a valid observable. It can be experimentally measured for systems such as this. But it is also very abstract and obtuse. A far more understandable quantity is the particle separation. It is also going to throw up some surprising results. The particle separation is given

$$\begin{aligned}
\langle r(t) \rangle &= \langle \psi_n(t) | r | \psi_n(t) \rangle, \\
&= \sum_{j,k=0}^{\infty} \langle \psi_n(0) | \phi_j \rangle \langle \phi_k | \psi_n(0) \rangle \langle \phi_j | r | \phi_k \rangle e^{-i(E_j - E_k)t/\hbar}.
\end{aligned} \tag{19}$$

Where the quench happens at $t = 0$, ϕ_k are eigenstates of the post-quench Hamiltonian with eigenenergy E_k . Again, you can choose whatever initial state you want but have no direct control

over the final state. The $\langle \psi_n(0) | \phi_j \rangle$ terms have already been considered in Section 4 leaving just the cross terms. We have two cases for the cross terms interacting and non-interacting. Let's start with the interacting case.

$$\begin{aligned}
\langle \phi_j | r | \phi_k \rangle &= \int \phi_{\nu_j} r \phi_{\nu_k} d^3r, \\
&= N_{\nu_j} N_{\nu_k} \int r e^{\frac{-r^2}{a_\mu^2}} \Gamma(-\nu_j) \Gamma(-\nu_k) U\left(-\nu_j, \frac{3}{2}, \frac{r^2}{a_\mu^2}\right) U\left(-\nu_k, \frac{3}{2}, \frac{r^2}{a_\mu^2}\right) d^3r, \\
&= \frac{2}{a_\mu^3 \sqrt{Z(\nu_j) Z(\nu_k)}} \int_{r=0}^{r=\infty} r^3 e^{\frac{-r^2}{a_\mu^2}} \Gamma(-\nu_j) U\left(-\nu_j, \frac{3}{2}, \frac{r^2}{a_\mu^2}\right) \Gamma(-\nu_k) U\left(-\nu_k, \frac{3}{2}, \frac{r^2}{a_\mu^2}\right) dr, \\
x &= \frac{r^2}{a_\mu^2}, \quad dr = \frac{dr}{dx} dx = \frac{a_\mu^2}{2r} dx, \\
&= \frac{a_\mu}{\sqrt{Z(\nu_j) Z(\nu_k)}} \int_{r=0}^{r=\infty} x e^{-x} \Gamma(-\nu_j) U\left(-\nu_j, \frac{3}{2}, x\right) \Gamma(-\nu_k) U\left(-\nu_k, \frac{3}{2}, x\right) dx.
\end{aligned}$$

Recall Eq. (5), you can expand Tricomi's function in terms of Laguerre polynomials. This gives

$$\langle \phi_j | r | \phi_k \rangle = \frac{a_\mu}{\sqrt{Z(\nu_j) Z(\nu_k)}} \sum_{m,n=0}^{\infty} \int_{x=0}^{x=\infty} x e^{-x} \left(\frac{L_m^{\frac{1}{2}}(x)}{j - \nu_j} \right) \left(\frac{L_n^{\frac{1}{2}}(x)}{k - \nu_k} \right) dx.$$

From Gradshteyn & Ryzhik (2014) we have the identity

$$\begin{aligned}
\int_0^\infty e^{-x} x^{\alpha+\beta} L_n^\alpha(x) L_m^\beta(x) dx &= (-1)^{n+m} (\alpha + \beta)! \binom{\alpha + n}{m} \binom{\beta + m}{n}, \\
\text{where } \operatorname{Re}[\alpha + \beta] &> -1.
\end{aligned} \tag{20}$$

We have $\alpha = \beta = 1/2$ so this applies. This leaves us with

$$\begin{aligned}
\langle \phi_j | r | \phi_k \rangle &= \frac{a_\mu}{\sqrt{Z(\nu_j) Z(\nu_k)}} \sum_{m,n=0}^{\infty} \frac{(-1)^{m+n}}{(m - \nu_j)(n - \nu_k)} \\
&\quad \times \binom{m + 1/2}{n} \binom{n + 1/2}{m}, \\
&= \frac{a_\mu}{\sqrt{Z(\nu_j) Z(\nu_k)}} \sum_{m,n=0}^{\infty} \frac{(-1)^{m+n}}{(m - \nu_j)(n - \nu_k)} \\
&\quad \times \frac{\Gamma(m + 3/2)}{\Gamma(n + 1) \Gamma(m - n + 3/2)} \frac{\Gamma(n + 3/2)}{\Gamma(m + 1) \Gamma(n - m + 3/2)}, \\
&= \frac{a_\mu}{\sqrt{Z(\nu_j) Z(\nu_k)}} \sum_{m,n=0}^{\infty} \frac{(-1)^{m+n}}{(m - \nu_j)(n - \nu_k)} \binom{m + 1/2}{n} \binom{n + 1/2}{m}.
\end{aligned} \tag{21}$$

In the unitary limit this becomes

$$\begin{aligned}
\langle \phi_j | r | \phi_k \rangle &= \frac{a_\mu}{\pi^2} \sqrt{\frac{\Gamma(j + 1/2) \Gamma(k + 1/2)}{\Gamma(j + 1) \Gamma(k + 1)}} \\
&\quad \times \sum_{m,n=0}^{\infty} \frac{(-1)^{m+n}}{(m - j + 1/2)(n - k + 1/2)} \binom{m + 1/2}{n} \binom{n + 1/2}{m}.
\end{aligned}$$

Luckily this actually has an exact form. If you feed this into mathematica and jiggle it a bit you get

$$\begin{aligned}\langle \tilde{r}(t) \rangle &= \frac{-4}{\pi^{5/2}} \sum_{m,l=0}^{\infty} \frac{(-1)^{m+l}}{1+2m} e^{-i\omega t(1+2l)} \beta(e^{-2i\omega t}, \frac{-1}{2} - l, \frac{3}{2}), \\ &\times \binom{\frac{1}{2} + m}{l} \binom{\frac{1}{2} + l}{m} {}_2F_1\left(\frac{-1}{2}, \frac{-1}{2} - m, \frac{1}{2} - m, e^{2i\omega t}\right),\end{aligned}$$

where $\beta(z, a, b)$ is the incomplete beta function:

$$\beta(z, a, b) = \int_0^z t^{a-1} (1-t)^{b-1} dt.$$

Now we move onto the reverse (unitary to non-interacting) quench

$$\begin{aligned}\langle \phi_m | r | \phi_n \rangle &= \int \phi_m r \phi_n d^3 \vec{r}, \\ &= \int \phi_m r \phi_n d^3 \vec{r}, \\ &= N_m N_n \int_0^\infty r^3 e^{-r^2/a_\mu^2} L_m^{\frac{1}{2}}\left(\frac{r^2}{a_\mu^2}\right) L_n^{\frac{1}{2}}\left(\frac{r^2}{a_\mu^2}\right) dr, \\ x &= \frac{r^2}{a_\mu^2}, \quad dr = \frac{dr}{dx} dx = \frac{a_\mu^2}{2r} dx, \\ \langle \phi_m | r | \phi_n \rangle &= \frac{a_\mu^4 N_m N_n}{2} \int_0^\infty \sqrt{x} e^{-x} L_m^{\frac{1}{2}}(x) L_n^{\frac{1}{2}}(x) dx, \\ &= a_\mu \sqrt{\frac{\Gamma(n+1)\Gamma(m+1)}{\Gamma(n+\frac{3}{2})\Gamma(m+\frac{3}{2})}} \int_0^\infty x e^{-x} L_m^{\frac{1}{2}}(x) L_n^{\frac{1}{2}}(x) dx.\end{aligned}$$

Recall the identity from Eq. (20),

$$\begin{aligned}&= a_\mu \sqrt{\frac{\Gamma(n+1)\Gamma(m+1)}{\Gamma(n+\frac{3}{2})\Gamma(m+\frac{3}{2})}} (-1)^{n+m} \binom{n+\frac{1}{2}}{m} \binom{m+\frac{1}{2}}{n} \\ \langle \phi_m | r | \phi_n \rangle &= \sqrt{\frac{\Gamma(m+\frac{3}{2})\Gamma(n+\frac{3}{2})}{\Gamma(m+1)\Gamma(n+1)}} \frac{a_\mu (-1)^{m+n}}{\Gamma(n-m+\frac{3}{2})\Gamma(m-n+\frac{3}{2})}\end{aligned}$$

Now to plot $\langle r(t) \rangle$ and see what happens.

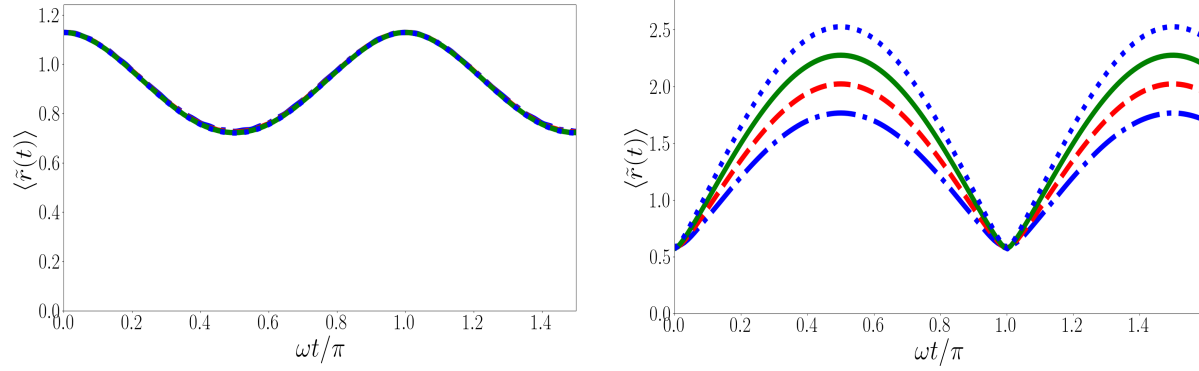


Figure 3: Expectation value of r as a function of time. First panel is the non-interacting to unitary quench, second is the unitary to non-interacting quench. In the first panel, the blue dotted curve corresponds to $N_{\max} = 160$, the green solid curve corresponds to $N_{\max} = 80$, the red dashed curve corresponds to $N_{\max} = 40$ and the dot-dashed blue curve corresponds to $N_{\max} = 20$. Because this sum is convergent all curves are overlaid on top of one another. In the second panel each curve corresponds to a different cut-off ($N_{\max}$) in the sum over the two inserted complete sets of state. Specifically, the dotted blue curve corresponds to $N_{\max} = 80$, the solid green curve corresponds to $N_{\max} = 40$, the dashed red curve corresponds to $N_{\max} = 20$ and the blue dot-dashed curve corresponds to $N_{\max} = 10$.

As we can see we get a divergence for the reverse quench. Notably, this happens for whatever quench so long as the final scattering length is zero, i.e. towards the non-interacting limit. We will see later that this is a result of something called the dynamic sweep theorem.

6 Energy Expectation

Given the particle separation is diverging something is clearly going wrong. It seems quite natural to us to look at what happens to the energy after the quench. The expectation of the total energy is

$$\begin{aligned}
 \langle \hat{E}(t) \rangle &= \langle \psi(t) | \hat{H} | \psi(t) \rangle = \sum_{j,k=0}^{\infty} \langle \psi(0) | \phi_j \rangle \langle \phi_j | \hat{H} | \phi_k \rangle \langle \phi_k | \psi(0) \rangle e^{-i(E_j - E_k)t/\hbar}, \\
 &= \sum_{j,k=0}^{\infty} \langle \psi(0) | \phi_j \rangle E_j \delta_{j,k} \langle \phi_k | \psi(0) \rangle e^{-i(E_j - E_k)t/\hbar}, \\
 \langle \hat{E}(t) \rangle &= \sum_{j=0}^{\infty} E_j |\langle \psi(0) | \phi_j \rangle|^2.
 \end{aligned} \tag{22}$$

Calculating this is quite straightforward. We already have the wavefunction overlaps and the eigenenergies.

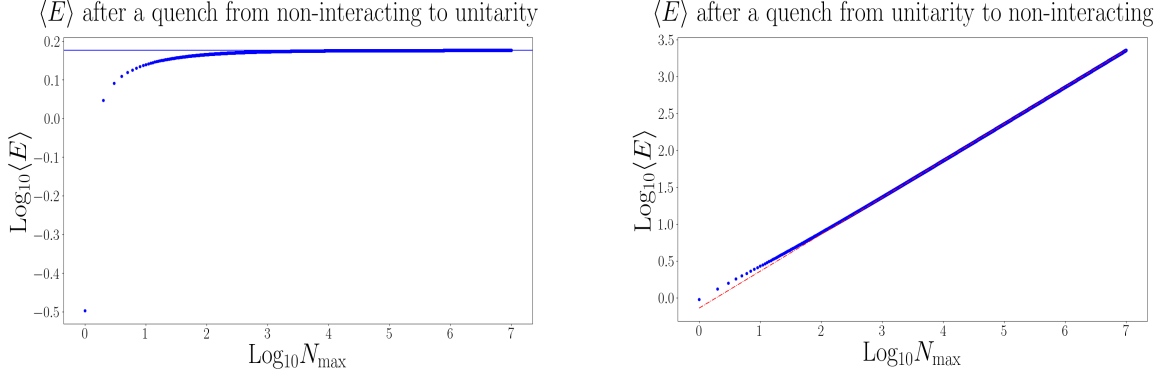


Figure 4: Expectation value of the energy as a function of the number of terms in the expansion in Eq. (22). First figure corresponds to the forwards quench and it converges nicely. The second figure corresponds to the reverse quench. It diverges. The red dot-dashed line is a line of best fit with gradient ≈ 0.48 (it's a logarithmic divergence, so linear on the log-log scale). This implies that the energy diverges as the square root of the number of terms.

The $\langle E \rangle$ is divergent for the to non-interacting quenches like with $\langle r \rangle$. This makes sense, the particles becoming far apart would imply a great deal of potential energy from the trap.

Worth noting that at such high values of $N_{\max}$ you start to get numeric issues due to the gamma functions. Often we have a gamma function divided by a gamma function. For large indices this leads to huge number divided by even bigger number to get a very small number. But the computer still needs to deal with those huge numbers in the first place so it gives up. You can get around this a by using the identity (from e.g. wikipedia)

$$\frac{\Gamma(n + \frac{1}{2})}{\Gamma(n + 1)} = \binom{n - \frac{1}{2}}{n} \sqrt{\pi}.$$

Also, for some reason I can't figure out, when you try to solve the transcendental for ν and you use the binomial coeff form you get weird stuff at the 170th and 171st coeffs if you use the numpy solver. Just those coeffs though, it agrees with Mathematica as otherwise.

7 Probability Distribution Evolution

For quenches to the non-interacting limit we have divergences. One clear avenue to investigate further is to look at the probability distribution of the particle separation and see how it evolves over time.

The particle separation distribution is Eq. (19) where in the cross term $\langle \phi_j | r | \phi_k \rangle$ the integral over r hasn't been done. In mathematical language

$$\begin{aligned} P(r, t) &= \int_{\Omega} \psi(t) r \psi(t) d\Omega, \\ &= \sum_{j,k=0}^{\infty} \langle \psi(0) | \phi_j \rangle \langle \phi_k | \psi(0) \rangle \int_{\Omega} \phi_j(r) r \phi_k(r) d\Omega. \end{aligned} \quad (23)$$

Where $d\Omega$ refers to integrating over solid angle. Everything needed to evaluate this has been presented above so we'll just move onto the figures.

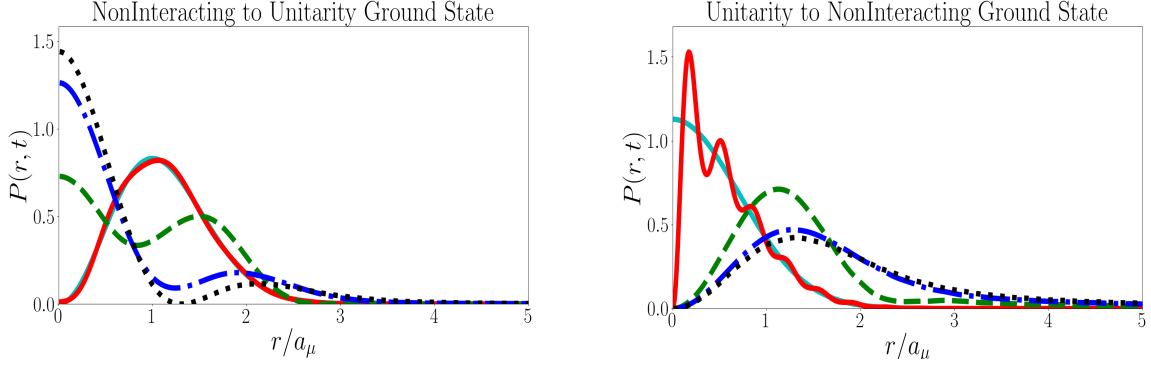


Figure 5: The evolution of $P(r, t)$. In the left hand plot we display the forwards quench, in the right hand plot we display the reverse quench. The solid cyan line corresponds to the static probability distribution at $t = 0$ for the initial state evaluated directly, i.e. not expanded over basis states. The solid red line is $P(r, t = 0)$ evaluated for 80 terms, i.e. an upper limit of 80 rather than infinity in the summation in Eq. (23). The agreement is good for the forwards quench but not for the reverse quench. The green dashed, blue dot-dashed, and black dotted lines correspond to $r = 0.17$, $t = 0.34$ and $t = 0.5$ respectively.

The reason for the divergence in the reverse quench is the long $1/r^2$ tail that forms away from $t = n\omega t$.

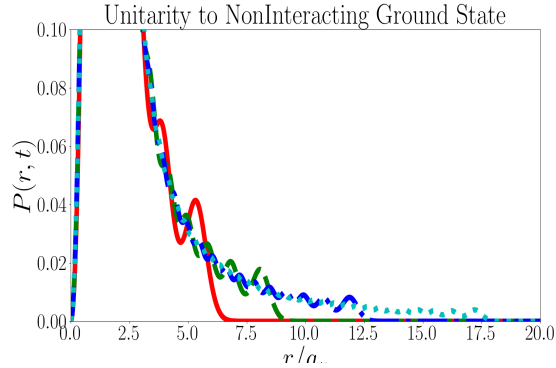


Figure 6: Close up of the tail of $P(r, t = 0.5)$ for various $N_{\max}$. The solid red, dashed green, dot-dashed blue and dotted can correspond to $N_{\max} = 4, 8, 16, 32$.

As you add more and more terms into Eq. (23) the tail gets longer and longer before ending in an exponential cut-off. This tail means that while $P(r, t = 0.5)$ is normalisable the first moment of $P(r, t = 0.5)$, the expectation of the particle separation, is divergent.

8 The Dynamic Sweep Theorem

The dynamic sweep theorem is a result of Shina Tan in Tan (2008) and it is highly relevant to quench dynamics. It relates the change in energy of a two-component fermion gas to a change in s -wave scattering length and external potential. It is stated

$$\frac{dE}{dt} = \frac{\hbar^2 \Omega C(t)}{8\pi\mu} \frac{d[-a_s(t)^{-1}]}{dt} + \sum_{\sigma} \int \rho_{\sigma}(\vec{r}t) \frac{\partial V_{\text{ext}}(\vec{r}t)}{\partial t} d^3r. \quad (24)$$

Here, E is the total energy, t is time, Ω is the volume of the system, σ refers to the spin state (fermion species) and V_{ext} is the trap potential. In our case the trap isn't changing and let's assume that the change in a_s follows a sigmoid type function.

$$a_s(t) = \frac{a_f - a_i}{\pi} \arctan(xt) + \frac{a_f + a_i}{2}. \quad (25)$$

Where x parametrises the speed of the quench.

$$\frac{d[-a_s(t)^{-1}]}{dt} = \frac{1}{a_s^2} \frac{da_s(t)}{dt} = \frac{x(a_f - a_i)}{\pi(1 + x^2 t^2) a_s(t)^2}.$$

Because we're assuming a rapid quench we have that the scattering length changes much more rapidly than the contact. That is, the contact is constant on the timescale we care about, i.e. $C(t) = C(0)$. Of course the contact does actually change but the quench is so rapid that compared to scattering length we can treat it as unchanging.

$$\begin{aligned} \int_{-t'}^{+t'} \frac{dE}{dt} dt &= \int_{-t'}^{+t'} \frac{\hbar^2 \Omega C(t)}{8\pi\mu} \frac{d[-a_s(t)^{-1}]}{dt} dt. \\ \Delta E &= \frac{\hbar^2 \Omega C(0)}{8\pi\mu} \int_{-t'}^{+t'} \frac{d[-a_s(t)^{-1}]}{dt} dt \\ &= \frac{\hbar^2 \Omega C(0)}{8\pi\mu} \left[\frac{-2\pi}{(a_f + a_i)\pi + 2(a_f - a_i) \arctan(xt)} \right]_{t=-t'}^{t=t'}. \end{aligned}$$

For a very fast quench we have $x \rightarrow \infty$.

$$\lim_{x \rightarrow \infty} \left[\frac{-2\pi}{(a_f + a_i)\pi + 2(a_f - a_i) \arctan(xt)} \right]_{t=-t'}^{t=t'} = \frac{-2t}{a_i(t - |t|) + a_f(t + |t|)}$$

Putting it together

$$\Delta E = \frac{\hbar^2 \Omega C(0)}{8\pi\mu} \left(\frac{1}{a_i} - \frac{1}{a_f} \right). \quad (26)$$

Now we have to derive the contact C . Again from Tan (2008)

$$C = \Omega^{-1} 16\pi^2 \int |A(\vec{r}_0)|^2 d^3 r_0 \quad (27)$$

where $A(\vec{r}_0)$ is defined as part of the short range condition on the wavefunction $\phi(\vec{r}_1, \vec{r}_2)$.

$$\phi(\vec{r}_1, \vec{r}_2) = A(\vec{r}_0)(1/r - 1/a_s) + \mathcal{O}(r).$$

where $\vec{r}_0 = (\vec{r}_1 + \vec{r}_2)/2$, the centre-of-mass coordinate. So for us in our framework $A(\vec{r}_0)$ is

$$A(\vec{r}_0) = \psi_{\text{COM}} \lim_{r \rightarrow 0} r \psi_{\text{rel}}.$$

So we have

$$C = \Omega^{-1} 16\pi^2 \left| \lim_{r \rightarrow 0} r \psi_{\text{rel}} \right|^2, \quad (28)$$

and this is the result of Enss et al. (2022). For the interacting wavefunction we have

$$\begin{aligned}
C\Omega &= 16\pi^2 \lim_{r \rightarrow 0} |r^2 N_\nu^2 \Gamma(-\nu)^2 e^{-r^2/a_\mu^2} U(-\nu, 3/2, r^2/a_\mu^2)|, \\
&= 16\pi^2 N_\nu^2 \Gamma(-\nu)^2 \lim_{r \rightarrow 0} |r^2 \left(\frac{\sqrt{\pi}}{\Gamma(-\nu)} \frac{a_\mu}{r} - \frac{2\sqrt{\pi}}{\Gamma(-\nu - \frac{1}{2})} + \mathcal{O}\left(\frac{r}{a_\mu}\right) \right)^2|, \\
&= 16\pi^3 a_\mu^2 N_\nu^2
\end{aligned} \tag{29}$$

For the non-interacting wavefunction we have

$$C\Omega = 16\pi^2 \lim_{r \rightarrow 0} |r^2 N_n^2 e^{r^2/a_\mu^2} L_n^{\frac{1}{2}}(r^2/a_\mu^2)| = 0. \tag{30}$$

Putting it together

$$\Delta E = \frac{2\pi^2 a_\mu^2 N_\nu^2 \hbar^2}{\mu} \left(\frac{1}{a_i} - \frac{1}{a_f} \right). \tag{31}$$

There's a subtlety with the initially non-interacting case. The contact is 0 but there's the $1/(a_i = 0)$ term. The interacting wavefunction approaches the non-interacting form as $\nu \rightarrow n \in \mathbb{Z}_{\geq 0}$ so let's consider the initially interacting case and look at the appropriate limit.

$$\begin{aligned}
\Delta E &= \frac{4\pi^2 a_\mu^2 N_\nu^2 \hbar^2}{\mu} \left(\frac{1}{a_i} - \frac{1}{a_f} \right). \\
&= \frac{4\pi^2 a_\mu^2 \hbar^2}{\mu} \left(2\pi^2 a_\mu^3 \frac{\Gamma(1-\nu) [\psi^{(0)}(-\nu - \frac{1}{2}) - \psi^{(0)}(-\nu)]}{\nu \Gamma(-\nu - \frac{1}{2})} \right)^{-1} \left(\frac{2\Gamma(-\nu)}{a_\mu \Gamma(-\nu - 1/2)} - \frac{1}{a_f} \right) \\
&= \frac{-4\hbar^2}{\pi \mu a_\mu^2 [\psi^{(0)}(\nu - \frac{1}{2}) - \psi^{(0)}(-\nu)]} - \frac{4\pi^2 a_\mu^2 N_\nu^2 \hbar^2}{\mu a_f}
\end{aligned}$$

For the non-interacting case $\nu = 0, 1, 2, 3$, which causes $\psi^{(0)}(-\nu) \rightarrow \infty$ so the denominator is infinity and the second term goes to zero since it's N_ν^2 times some constants. So $\Delta E = 0$. Essentially the contact “goes to zero” faster than the $1/a_i$ term so as a whole $\Delta E = 0$.

Now it's useful to consider the question of if the divergence is caused by the instantaneous quench or the zero-range interaction, or both. Recall

$$\begin{aligned}
a_s(t) &= \frac{a_f - a_i}{\pi} \arctan(xt) + \frac{a_f + a_i}{2}. \\
\Delta E &= \frac{\hbar^2 \Omega C(0)}{8\pi \mu} \left[\frac{-2\pi}{(a_f + a_i)\pi + 2(a_f - a_i) \arctan(xt)} \right]_{t=-t'}^{t=t'}.
\end{aligned}$$

Holding off from doing the $x \rightarrow \infty$ limit for now.

$$\Delta E = \frac{\hbar^2 \Omega C(0)}{8\pi \mu} \left[\frac{8\pi(a_f - a_i) \arctan(xt')}{\pi^2(a_f + a_i)^2 - 4(a_f - a_i)^2 \arctan(xt')^2} \right] \tag{32}$$

In this case we have a finite ΔE even if $a_f = 0$. It's only if we have an instantaneous quench ($x \rightarrow \infty$) do we have a divergence issue. So we do need the quench to be instantaneous to get a divergence when we have zero range interactions.

9 A Collection of Useful Results

The purpose of this subsection is to have one place with all the results. The working and context for them is all below but it is useful just to have everything in one place so you aren't constantly looking for that one thing. This section will get expanded over time

The interacting wavefunction

$$\begin{aligned}\psi_{2b}(\nu, r) &= N_\nu \Gamma(-\nu) e^{\frac{r^2}{2a_\mu^2}} U\left(-\nu, \frac{3}{2}, \frac{r^2}{a_\mu^2}\right), \\ N_\nu &= (2\pi a_\mu^3 Z(\nu))^{-\frac{1}{2}}, \\ Z(\nu) &= \frac{\pi \Gamma(1-\nu) (\psi^{(0)}(-\nu - \frac{1}{2}) - \psi^{(0)}(-\nu))}{\nu \Gamma(-\nu - \frac{1}{2})}.\end{aligned}$$

In the unitary limit

$$\begin{aligned}Z(\nu_j) &= \pi^2 \frac{\Gamma(1+j)}{\Gamma(\frac{1}{2}+j)}, \\ N_{\nu_j} &= \sqrt{\frac{\Gamma(\frac{1}{2}+j)}{2\pi^3 a_\mu^3 \Gamma(1+j)}} = \sqrt{\frac{1}{2\pi^{5/2} a_\mu^2} \binom{j - \frac{1}{2}}{j}}.\end{aligned}$$

The non-interacting wavefunction(for $l = 0$)

$$\begin{aligned}\phi_n(r) &= N_n e^{\frac{-r^2}{2a_\mu^2}} L_n^{1/2}\left(\frac{r^2}{a_\mu^2}\right) Y_{00}(\hat{r}), \\ N_n &= \sqrt{\sqrt{\frac{1}{4\pi a_\mu^2} \frac{2^{n+3} \Gamma(n+1)}{\Gamma(n + \frac{3}{2})}}}, \\ &= \sqrt{\frac{2\Gamma(n+1)}{a_\mu^3 \Gamma(n + \frac{3}{2})}} = \sqrt{\frac{2}{a_\mu^3 \sqrt{\pi} (n + \frac{1}{2})} \binom{n - \frac{1}{2}}{n}^{-1}}\end{aligned}$$

Eigenenergies are given by the transcendental

$$\frac{a_\mu}{a_s} = \frac{2\Gamma(-\nu)}{\Gamma(-\nu - 1/2)} = \frac{2}{\sqrt{\pi} \binom{-\nu - \frac{3}{2}}{-\nu - 1}}$$

Overlap of two interacting wavefunctions.

$$\langle \phi_{\nu_i} | \phi_{\nu_j} \rangle = \frac{\sqrt{\pi}}{2\nu_i \nu_j} \frac{1}{\sqrt{Z(\nu_i) Z(\nu_j)}} {}_3F_2\left(\frac{3}{2}, -\nu_j, -\nu_i; 1 - \nu_j, 1 - \nu_i; 1\right)$$

Non-interacting-interacting wavefunction overlap. Non-interacting index is n , interacting index is j .

$$\langle \psi_n | \phi_{\nu_j} \rangle = \frac{1}{(n - \nu_j)} \sqrt{\frac{\Gamma(n + \frac{3}{2})}{\Gamma(n + 1)}} (Z(\nu_j))^{-1/2}$$

Non-interacting-unitary wavefunction overlap. Non-interacting index is n , unitary index is j

$$\begin{aligned}\langle \psi_n | \phi_{\nu_j} \rangle &= \frac{\sqrt{n + \frac{1}{2}}}{\pi(n - j + \frac{1}{2})} \sqrt{\frac{\Gamma(n + \frac{1}{2})\Gamma(j + \frac{1}{2})}{\Gamma(n + 1)\Gamma(j + 1)}} \\ &= \frac{\sqrt{n + \frac{1}{2}}}{\sqrt{\pi}(n - j + \frac{1}{2})} \sqrt{\binom{n - \frac{1}{2}}{n} \binom{j - \frac{1}{2}}{j}}\end{aligned}$$

Cross term for the expectation value of r for a to interacting quench.

$$\langle \phi_j | r | \phi_k \rangle = \frac{a_\mu}{\sqrt{Z(\nu_j)Z(\nu_k)}} \sum_{m,n=0}^{\infty} \frac{(-1)^{m+n}}{(m - \nu_j)(n - \nu_k)} \binom{m + 1/2}{n} \binom{n + 1/2}{m}.$$

In the unitary limit this becomes

$$\begin{aligned}\langle \phi_j | r | \phi_k \rangle &= \frac{a_\mu}{\pi^2} \sqrt{\frac{\Gamma(j + 1/2)\Gamma(k + 1/2)}{\Gamma(j + 1)\Gamma(k + 1)}} \\ &\times \sum_{m,n=0}^{\infty} \frac{(-1)^{m+n}}{(m - j + 1/2)(n - k + 1/2)} \binom{m + 1/2}{n} \binom{n + 1/2}{m}.\end{aligned}$$

Cross term for the expectation value of r for a to non-interacting quench.

$$\langle \phi_m | r | \phi_n \rangle = \sqrt{\frac{\Gamma(m + \frac{3}{2})\Gamma(n + \frac{3}{2})}{\Gamma(m + 1)\Gamma(n + 1)}} \frac{a_\mu(-1)^{m+n}}{\Gamma(n - m + \frac{3}{2})\Gamma(m - n + \frac{3}{2})}$$

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